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Supplementary Materials for

Synthesis of stable and low-CO_2 selective ε-iron carbide Fischer-Tropsch catalysts

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Supplementary Text

Section S1. Mass balance calculation

Gasification stage

A. $C + H_2O \rightarrow H_2 + CO$

B. $C + ½ O_2 \rightarrow CO$

Water-gas shift stage

C. $CO + H_2O \rightarrow CO_2 + H_2$

Fischer-Tropsch synthesis stage

D. $2 CO + H_2 \rightarrow CO_2 + CH_2$

E. $CO + 2 H_2 \rightarrow H_2O + CH_2$

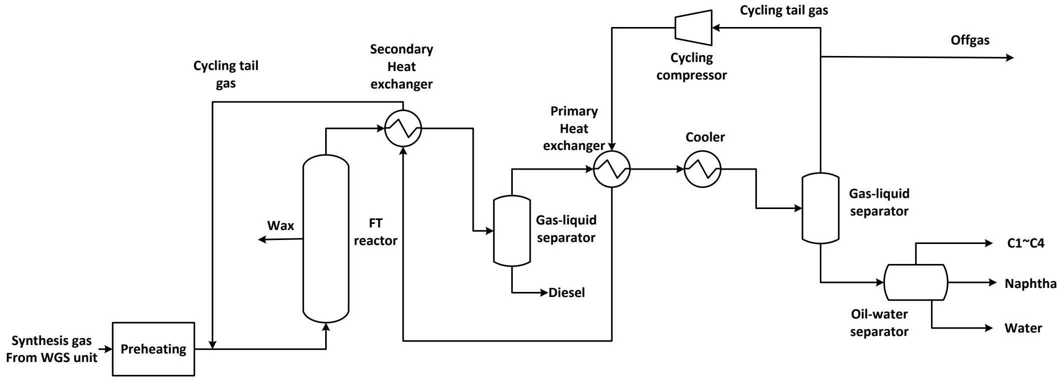
In the gasification unit, reactions A and B proceed in a ratio of $\frac{a}{1-a}$ ($0 < a < 1$) and we set the total C consumption at unity. The synthesis gas produced in the gasification unit then undergoes the water-gas shift (WGS) reaction with an extent of c (*e.g.*, c = 0.5 means that half of the CO is converted to CO_2 in the WGS unit). Thereafter, the synthesis gas fed to the FT unit is converted to C_nH_{2n}, H_2O and CO_2. The ratio of reactions D and E is determined by the H_2/CO ratio of the synthesis gas, *i.e.*, the CO_2 selectivity in FT is a function of the H_2/CO ratio of the synthesis gas entering the FT unit. Mass balance calculation yields the following:

1) total H_2O input (gasification + WGS) is $c + a$; net H_2O consumption (gasification + WGS – FT) is $\frac{1+a}{3}$;
2) Total CH_2 production (FT) is $\frac{1+a}{3}$
3) Total CO_2 release (WGS + FT) is $\frac{2-a}{3}$
4) Total O input (O_2 and H_2O) is $\frac{4-2a}{3}$

Based on these considerations, we can draw the following intermediate conclusions:

1) Total CO_2 release depends on the ratio of the gasification reactions A and B, but is independent on the extent of the WGS reaction (reaction C);
2) The total O input equals the O removed by CO_2;
3) The extent of the WGS governs the ratio of the CO_2 released in WGS and FT;
4) Overall CO_2 selectivity is between 1/3 and 2/3, depending of the ratio of reaction A and B;
5) Net H_2O consumption equals the total CH_2 production.

Section S2. Energy consumption calculation



Scheme S1. Model of an FT plant. Synthesis gas after the WGS unit is pre-heated prior to entering FT reactor. The products are sequentially separated by a first gas-liquid separator, a second gas-liquid separator, and an oil-water separator to finally achieve diesel, naphtha, C_1-C_4 and water fractions. The tail gas is divided into two parts. Most of the tail gas (depending on the feed) is cycled back to the FT reactor to achieve high CO conversion, while the rest is released as off-gas.

The energy consumption of the FT plant as a function of CO_2 selectivity in the FT reaction is calculated using *Aspen Plus* 7.2. The calculations are done for a 500 kton a^{-1} oil product industrial FT plant (shown in scheme S1). The molar rate entering the WGS unit is 15625±1 kmol h^{-1} (CO + H_2) with a typical gasification output H_2/CO ratio of 0.67 (*37*). Afterwards, the H_2/CO ratio of the synthesis gas is adjusted by the WGS unit to the desired value that results in nearly complete synthesis gas conversion at a given CO_2 selectivity. This feed (heated to 190 ^oC after the WGS step) is used as input for this modelling. The results are listed in table S1. The following assumptions are employed:

1) The CO_2 selectivity is varied by changing the ratio of reaction D and E in the FT unit. When synthesis gas is exclusively converted via reaction D, the CO_2 selectivity is 50%; when converted via reaction E, CO_2 selectivity is zero. Therefore, a combination of reactions D and E results in a CO_2 selectivity between 0 and 50%.
2) One-pass CO conversion in the FT reactor (V = 1979 m^3) is 60%;
3) Reaction conditions: 235 ^oC, 23 bar;
4) The overall CO conversion in the FT reactor is 96.4±0.15 %, which is obtained by varying the recycle ratio of the tail gas;
5) The C_1-C_4 (molar) selectivity is 7%;

6) Catalyst deactivation is not taken into account;
7) The energy efficiency is assumed to be 100% for the heater and 80% for the compressor.

Section S3. Operando Mössbauer spectroscopy

The iron carbides formed in the FT synthesis can be classified according to the sites occupied by the carbon atoms in structures with carbon in octahedral interstices (e-Fe_2C, ε’-$Fe_{2.2}C$, e-Fe_3C) and structures with carbon in trigonal prismatic interstices (χ-Fe_5C_2, Fe_7C_3, θ-Fe_3C).

De Smit et al. showed that the e-Fe_3C phase is thermodynamically unstable with a high deformation energy and, therefore, is not likely to form under typical FTS conditions (*1*). Maksimov *et al*., using Mössbauer spectroscopy, proposed that e-Fe_2C has three Fe sites with an atomic ratio of *I* : *II* : *III* = 4 : 1.6 : 1 and respective room-temperature hyperfine fields of 17 ± 0.3, 23.7 ± 0.3, and 13 ± 0.6 T (*38*). Later, Raupp *et al*. and Caer *et al*. indicated that the presence of site *III* in e-Fe_2C has never been convincingly demonstrated and its presence is not essential to the analysis of the Mössbauer spectra (*28*, *39*). Therefore, we fitted the Mössbauer spectra of possible e-Fe_2C contributions with two sites having a site ratio of about 4:1.6.

The Mössbauer signal of site *I* of e-Fe_2C is similar and overlaps the Mössbauer signal of e'-$Fe_{2.2}C$ (17.3 T at room temperature), which is understandable as the two carbides have identical space groups [$P6_3$/mmc (194)] and identical lattice parameters (only the carbon concentration differs). To quantify the amount of the □o q_2C present in our spectra, we follow the assignment of Raupp *et al*. (*39*), multiplying the independently evaluated site *II* of e-Fe_2C by a factor of 3.5. The ε’-$Fe_{2.2}C$ amount is then determined by subtracting the calculated site *I* of e-Fe_2C from the overlapping Mössbauer signal of the two carbides.

The fitting parameters of the Mössbauer spectra of R-Fe, Fe/SiO_2 and insufficiently reduced R-Fe catalyst are listed in table S2, S3 and S4, respectively.

Table S1. FT plant modeling as a function of CO_2 selectivity in FT reactor.

CO_2 selectivity (%)	0	8	15	20	25	35	40	50
H_2/CO inlet ratio of FT reactor	2.00	1.76	1.55	1.40	1.25	0.95	0.80	0.50
Inlet CO (kmol h^{-1})	5218.8	5660.9	6126.6	6510.9	6943.8	8012.5	8681.3	10406.3
Inlet H_2 (kmol h^{-1})	10406.3	9964.1	9498.4	9114.1	8681.3	7612.5	6943.8	5218.8
CO in offgas (kmol h^{-1})	179.5	202.6	217.6	232.4	245.4	301.9	306.8	387.4
H_2 in offgas (kmol h^{-1})	331.8	358.7	340.6	325.2	309.0	288.2	244.8	209.5
Cycle ratio	0.8	1.3	1.8	2.2	2.6	3.8	4.4	6.0
Compressor power (kw)	2719.3	4258.4	5797.1	7110.7	8497.8	11993.9	13912.0	18862.1
Heating power (kw)	78014.9	81252.2	85249.1	88436.4	91662.8	100465.0	103548.5	107771.5
CO_2 production (kmol h^{-1})	0.0	408.2	842.5	1235.3	1617.0	2764.7	3158.6	4943.4
C_{5+} production (kmol h^{-1})*	4619.3	4575.8	4558.4	4538.7	4534.4	4549.9	4540.1	4426.0

* based on carbon number

Table S2. Fitting parameters of the Mössbauer spectra of R-Fe catalyst

(corresponding to Fig. 3)

Sample/ Treatment	IS (mm·s $^{-1}$)	QS (mm·s $^{-1}$)	Hyperfine field (T)	Γ (mm·s $^{-1}$)	Site	Site contribution (%)	Carbide	Carbide contribution(%)
Raney Fe carburized	0.28 0.24	- -	18.7 25.3	0.56 0.86	I II	89 11	ε -Fe_2C ε’-$Fe_{2.2}C$	39 61
Raney Fe H_2/CO=1.5 235 °C, 23 bar, 12 h	0.26 0.22	- -	18.8 25.2	0.68 0.68	I II	88 12	ε -Fe_2C ε’-$Fe_{2.2}C$	42 58
Raney Fe H_2/CO=1.5 saturator 250 °C, 23 bar, 12 h	0.28 0.23	- -	18.8 25.4	0.57 0.57	I II	92 8	ε -Fe_2C ε’-$Fe_{2.2}C$	28 72

Experimental uncertainties: Isomer shift: I.S. ± 0.02 mm s^{-1}; Quadrupole splitting: Q.S. ± 0.02 mm s^{-1}; Line width: Γ ± 0.03 mm s^{-1}; Hyperfine field: ± 0.1 T; Spectral contribution: ± 3%.

Table S3. Fitting parameters of the Mössbauer spectra of Fe/SiO_2 catalyst

(corresponding to fig. S2)

Sample/ Treatment	IS ($mm\cdot s^{-1}$)	QS ($mm\cdot s^{-1}$)	Hyper fine field (T)	Γ ($mm\cdot s^{-1}$)	Site	Site contribution (%)	Metal/ Carbide	Carbide Contribution(%)
Fe/SiO_2 catalysts carburized	0.00 0.25	- -	34.0 11.7	0.41 0.86	- -	- -	Fe^0 Fe_xC	39 61
Fe/SiO_2 catalysts H_2/CO=1.5 235 °C, 23 bar, 12 h	0.24 0.25	- -	18.6 24.0	0.54 0.57	I II	72 28	ε -Fe_2C ε'-$Fe_{2.2}C$	98 2
Fe/SiO_2 catalysts H_2/CO=1.5 saturator 250 °C, 23 bar, 12 h	0.25 0.25	- -	18.9 24.8	0.58 0.58	I II	73 27	ε -Fe_2C ε'-$Fe_{2.2}C$	94.5 5.5

Experimental uncertainties: Isomer shift: I.S. ± 0.02 mm s^{-1}; Quadrupole splitting: Q.S. ± 0.02 mm s^{-1}; Line width: Γ ± 0.03 mm s^{-1}; Hyperfine field: ± 0.1 T; Spectral contribution: ± 3%.

Table S4. Fitting parameters of the Mössbauer spectra of insufficiently reduced R-Fe catalysts (corresponding to fig. S5)

Sample/ Treatment	IS ($mm\cdot s^{-1}$)	QS ($mm\cdot s^{-1}$)	Hyperfine field (T)	Γ ($mm\cdot s^{-1}$)	Site	Spectral contribution (%)	Phase	Phase contribution(%)
Raney Fe Insufficiently Reduced by H_2 275 °C, 1 h Pretreatment H_2/CO=2 170 °C, 1 bar, 40 min	0.00 0.34 0.38	- -0.01 0.00	34.1 50.9 47.2	0.35 0.65 0.65	Fe^0 Fe^{3+} Fe^{3+}	64 24 12	Fe Fe^{3+}	64 36
Raney Fe Carburized	0.25 0.22 0.21 0.20 0.28 0.38 0.72	- - - - 0.02 -0.45 -0.04	18.6 25.4 20.4 12.7 50.7 45.7 46.5	0.46 0.61 0.61 0.61 0.77 0.77 0.77	ε'-$Fe_{2.2}C$ χ-Fe_5C_2 (I) χ-Fe_5C_2 (II) χ-Fe_5C_2 (III) Fe^{3+} Fe^{3+} Fe^{2+}	37 14 15 6 16 7 5	ε'-$Fe_{2.2}C$ χ-Fe_5C_2 Fe^{2+} Fe^{3+}	37 35 5 23
Raney Fe H_2/CO=1.5 saturator 235 °C, 23 bar, 24 h	0.24 0.25 0.19 0.22 0.31 0.70	- - - - -0.04 -0.65	18.6 25.2 20.8 13.0 51.3 48.5	0.45 0.62 0.62 0.62 0.63 0.63	ε'-$Fe_{2.2}C$ χ-Fe_5C_2 (I) χ-Fe_5C_2 (II) χ-Fe_5C_2 (III) Fe^{3+} Fe^{2+}	47 15 14 7 12 5	ε'-$Fe_{2.2}C$ χ-Fe_5C_2 Fe^{2+} Fe^{3+}	47 36 5 12
Raney Fe H_2/CO=1.5 saturator 250 °C, 23 bar, 12 h	0.25 0.21 0.19 0.22 0.32 0.48 0.70	- - - - 0.02 -0.29 -0.70	18.6 25.6 20.5 12.0 50.9 44.9 48.8	0.43 0.55 0.55 0.55 0.64 0.64 0.64	ε'-$Fe_{2.2}C$ χ-Fe_5C_2 (I) χ-Fe_5C_2 (II) χ-Fe_5C_2 (III) Fe^{3+} Fe^{3+} Fe^{2+}	38 15 15 7 13 5 7	ε'-$Fe_{2.2}C$ χ-Fe_5C_2 Fe^{2+} Fe^{3+}	38 37 7 18

Experimental uncertainties: Isomer shift: I.S. ± 0.02 mm s^{-1}; Quadrupole splitting: Q.S. ± 0.02 mm s^{-1}; Line width: Γ ± 0.03 mm s^{-1}; Hyperfine field: ± 0.1 T; Spectral contribution: ± 3%.

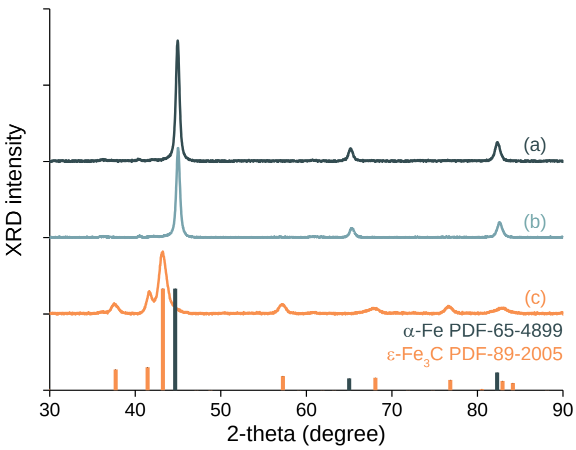


Fig. S1. In situ XRD patterns of the Fe/SiO_2 catalyst throughout the ε(′)-carbide synthesis procedure. (a) Reduction: H_2 flow, 1 bar, 430 ^oC, 1 h. (b) Pre-treatment: H_2/CO = 2, 1 bar, 170 ^oC, 40 min. (c) Carburization: H_2/CO = 1.5, 1 bar, ramping from 170 to 250 ^oC at a rate of 0.5 ^oC min^{-1}, kept for 1 h.

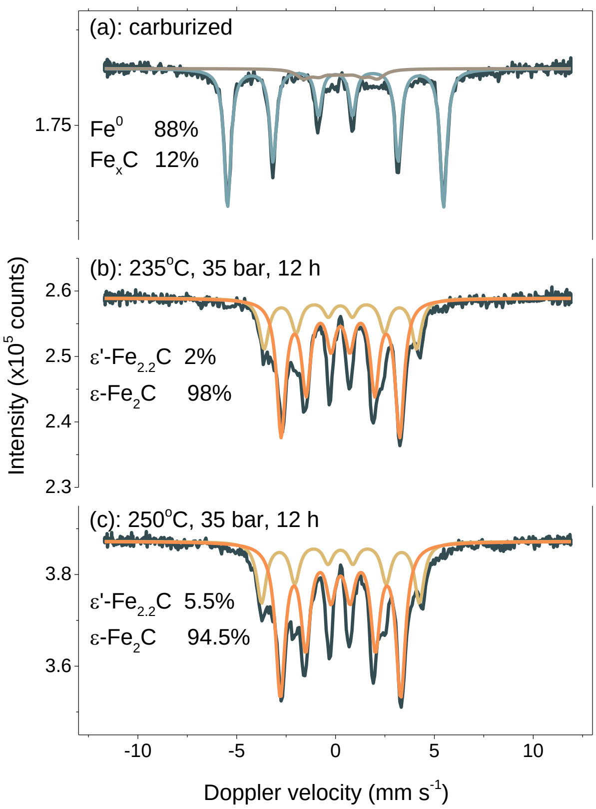


Fig. S2. Operando Mössbauer spectra of Fe/SiO_2 catalyst. (a) In situ reduced at 430 °C for 24 h followed by a pre-treatment for 40 min at the condition of H_2/CO = 2, 1 bar, 170 °C, and then ramped to 250 °C at a rate of 0.5 °C min^{-1} and kept for 1 h; (b) the sample was subsequently exposed to the condition of H_2/CO = 1.5, 23 bar, 235 °C and kept for 12 h; (c) afterward switched to the condition of H_2/CO = 1.5, saturated by vapor at room temperature, 23 bar, 250 °C and kept for 12 h. The data of (a), (b) were acquired at 120 K, (c) were acquired at 4.2 K.

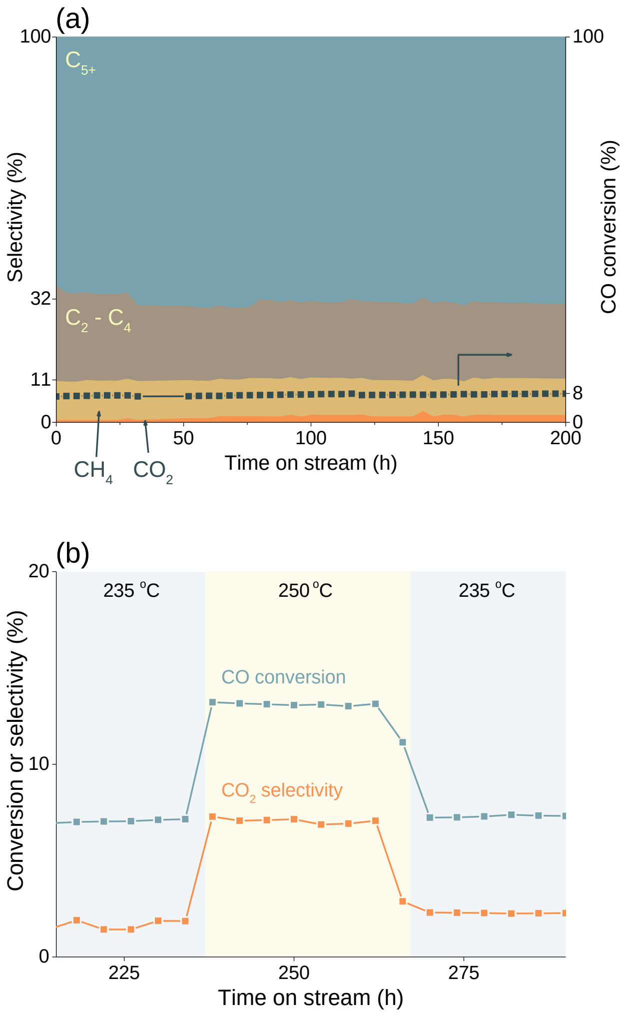


Fig. S3. FT performance of Fe/SiO_2 catalyst containing only ε(′)-carbide as a function of time on stream. (a) CO conversion and product distribution at H_2/CO = 1.5, 23 bar, 235 ^oC, GHSV = 18000 h^{-1}. (b) CO conversion and CO_2 selectivity during a temperature excursion to 250 ^oC for 50 h.

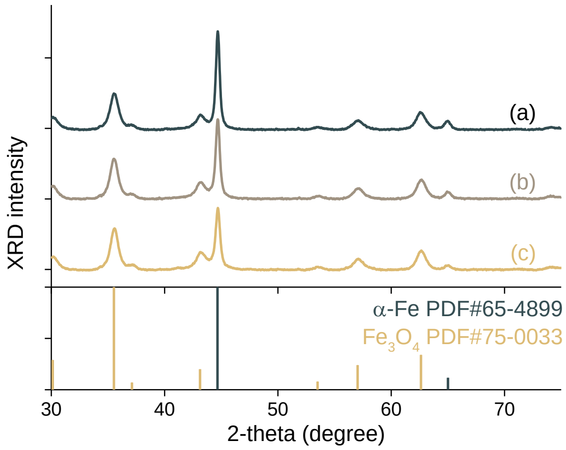


Fig. S4. In situ XRD patterns of the R-Fe catalyst throughout the ε(′)-carbide formation. (a) Reduction: H_2 flow, at 1 bar, 275 °C, 1 h. Notably this condition results in an insufficient reduction. (b) Pre-treatment: H_2/CO = 2, 1 bar, 170 °C, 40 min. (c) Carburization: H_2/CO = 1.5, 1 bar, ramping from 170 to 250 °C at a rate of 5 °C min^{-1}, kept for 60 min.

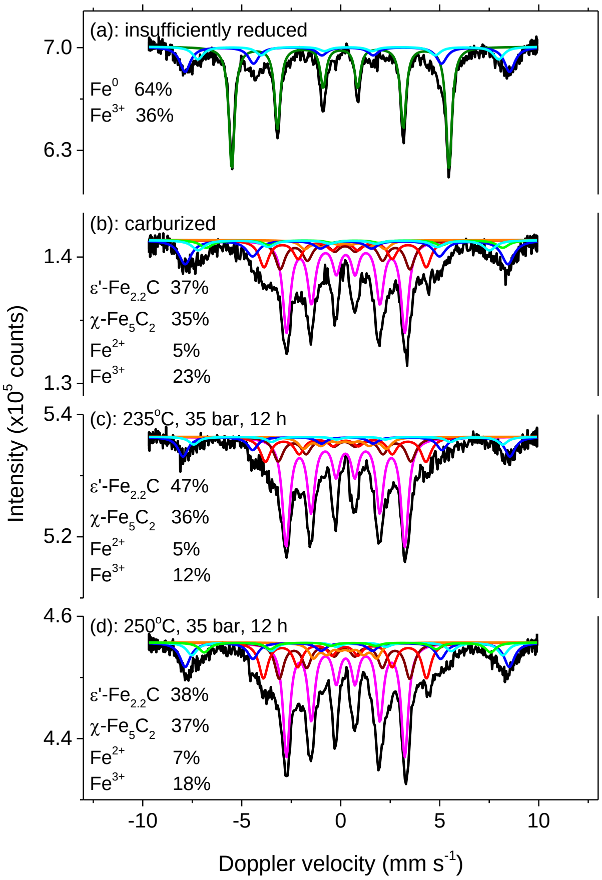


Fig. S5. Operando Mössbauer spectra of insufficiently reduced R-Fe catalyst recorded in different stages. (a) Reduced in H_2 flow, at 1 bar, 275 ^{o}C for 1 h. Notably the sample was not fully reduced at this condition. (b) The sample was carburized at the condition of H_2/CO = 2, 1 bar, 170 ^{o}C and kept for 40 min followed by carburization at a condition of H_2/CO = 1.5, 1 bar, 250 ^{o}C, kept for 1 h. The data were acquired at 4.2K.

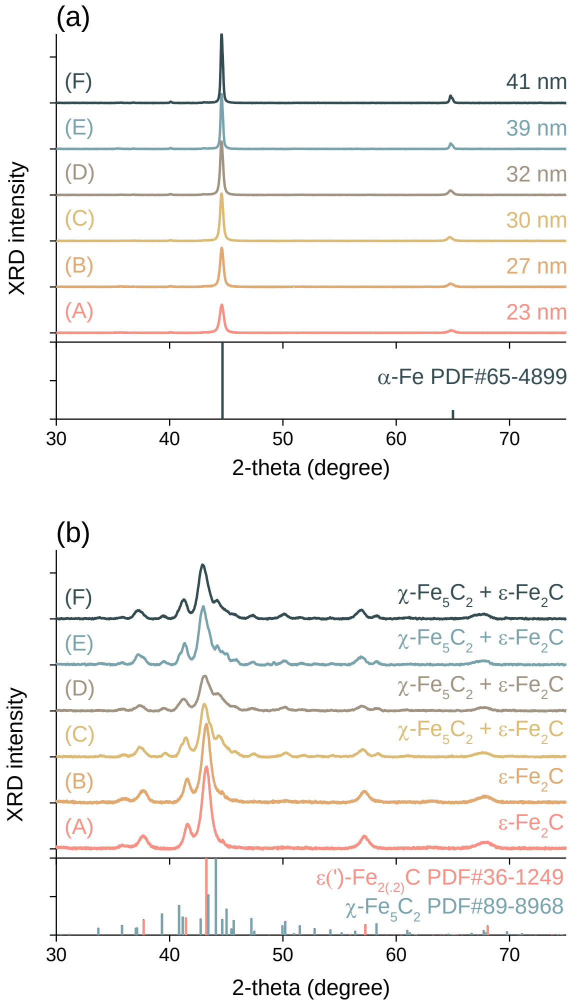


Fig. S6. In situ XRD patterns of various samples sequentially treated by reduction and carburization. In situ XRD patterns of (a) reduced and (b) carburized samples: (A) R-Fe reduced at 275 ^{o}C for 2 h; (B) R-Fe reduced at 430 ^{o}C for 1 h; (C) R-Fe reduced at 430 ^{o}C for 6 h; (D) R-Fe reduced at 600 ^{o}C for 1 h; (E) Sample prepared by precipitation, reduced at 375 ^{o}C for 2 h; (F) Sample prepared by precipitation, reduced at 430 ^{o}C for 2 h.

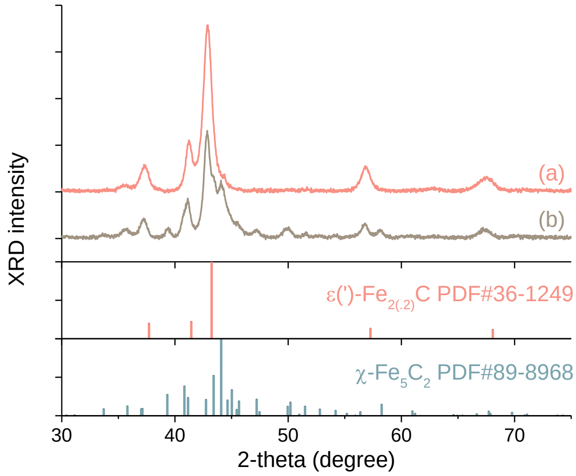


Fig. S7. In situ XRD patterns of the R-Fe catalysts synthesized following different procedures. (a) The catalyst was synthesized following (1) reduction: H_2 flow, 1 bar, 430 ^oC, 1 h; (2) pre-treatment: H_2/CO = 2, 1 bar, 170 ^oC, 40 min; (3) carburization: H_2/CO = 1.5, 1 bar, ramping from 170 to 250 ^oC at a rate of 0.5 ^oC min^{-1}, kept for 1 h. (b) The catalyst was synthesized following (1) reduction: H_2 flow, 1 bar, 430 ^oC, 1 h; (2) carburization: H_2/CO = 1.5, 1 bar, ramping from 50 to 250 ^oC at a rate of 0.5 ^oC min^{-1}, kept for 1 h.